EDUC-244 Report: Balancing Health and Taste

Introduction

We all live in a fast-moving world. In an attempt to do everything else, our health often takes a backseat. The general public is presented with an unlimited number of choices, stacked in grocery stores and at our fingertips. To prioritise health or to prioritise taste and convenience has indeed become a difficult choice for consumers. On the one hand, public health organisations such as the World Health Organisation and the Centres for Disease Control and Prevention highly advocate for better nutrition to combat mass health challenges such as Diabetes and Obesity. On the other hand, a common man at the end of a long day often craves 'comfort food'. The mission of our project is to investigate this issue using the food.com dataset. We are also through this project aiming to understand if the keywords in healthy recipes do not match with the keywords a user is searching for, such as comfort, cheesy, etc. And is it because of the vocabulary that healthy recipes often take a backseat when we search for food recipes online? And if that is indeed the case, can we build a recommendation system that would nudge the users towards healthy recipes that have the comforting keywords and also aren't too complex to make?

Research Questions:

At the start of this project, when we looked at the dataset, there were two burning questions that we discussed, which guided most of our analysis:

1. **The Semantic Gap**: Does the vocabulary used in recipes labelled Healthy lack the keywords such as rich, comfort, and creamy that drive user engagement? Does this, therefore, create a marketing disadvantage that makes healthy food seem less appealing and take a backseat?
2. **Feasibility of Auto-Recommendation**: Can we use data from recipes and user reviews to recommend recipes to users of food.com and similar websites? Can we nudge those users toward health without increasing the complexity of cooking a meal?

Stakeholder Analysis:

We identified 3 key stakeholders that would immensely benefit from our insights:

1. **Grocery Retailers such as Walmart & Instacart**: Their goal is to maximise revenue and customer retention. The insights from our semantic analysis can be integrated into their search categories, thus driving user engagement. They would inadvertently help in creating a larger impact by promoting healthy ingredients on their website/app.
2. **Public Health Organisations, such as WHO & CDC,** require data-driven insights to design better interventions and not just depend on nutritional labelling done on the back of the products.
3. **Users of Food.com & similar websites:** They visit the platform looking for recipes that are tasty and convenient. Health is often just a "good to have" feature.

Dataset

Source: Kaggle - Food.com Recipes & Interactions

Data Preprocessing: The raw dataset contains 230K+ recipes. It required a lot of cleaning and preprocessing before we could deep dive into analysing everything. The nutrition column was written as a list of values. We split this list into seven distinct columns: Calories #, Total Fat (PDV), Sugar (PDV), Sodium (PDV), Protein (PDV), Saturated Fat (PDV), and Carbohydrates (PDV). This granular breakdown was extremely important for analysing the different health metrics.

Feature Engineering: To answer our research questions, we engineered several new features:

1. **is_healthy Label:** A recipe was classified as *Healthy* if it met low thresholds for Sugar, Sodium, and Saturated Fat. Recipes exceeding the maximum PDV limits, such as >20% Daily Value, were labelled *Unhealthy*. And all the rest of the recipes, as they didn't really fit into a healthy or unhealthy category, were labelled as *Neutral*.
2. **Protein:** We calculated the ratio of Protein to Calories to measure nutrient density.
3. **Semantics:** We used text mining to flag search keywords such as *comfort* and *creamy* to quantify which vocabulary is used more for which health category.
4. **Ingredient Vectorisation:** To make the ingredient lists usable for modelling, we cleaned and tokenised the text data. We then selected the most common 300+ ingredients to reduce dimensionality and applied One-Hot Encoding. This created binary presence columns such as "egg" = 1 and “butter" = 0, which, combined with standardised calorie counts, formed our final feature matrix.

Questions: Upon looking at the dataset itself after doing the cleaning we discussed and brainstormed a lot of questions. Based on those questions, we decided which Machine Learning models would best suit to find the answers. Below are the questions we started to explore:

- What is the distribution of Key variables such as *calories*, *protein* and *time to cook*?
- What is the no. of Healthy and Unhealthy recipes?
- What are the top ingredients that people put in their meals?
- Do healthy recipes take longer to cook?
- Are healthy recipes more complicated to cook?
- Do users rate healthy food lower?
- Is there a semantic gap in the vocabulary used for healthy and unhealthy recipes?
- Which ingredients correlate with which nutritional features?
- Which recipes are clustered with recipes that users like?
- Do users actually rate recipes based on their taste preferences?
- Can we determine the quality of a recipe from this dataset?
- How do we nudge a user to eat healthy?

Methodology

Exploratory Data Analysis & Feature Engineering:

- To predict whether a recipe is Healthy, Neutral, or Unhealthy and identify which specific ingredients are most correlated with nutritional features
- Cleaned and tokenised the ingredient lists, selecting the most common 300+ ingredients. Applied One-Hot Encoding to create binary presence columns (e.g., egg = 1, butter = 0).
- The binary vectors were combined with features such as Calories, Cooking Time, and Number of steps to create the final feature matrix.

Multiclass Logistic Regression:

- Sampled 80,000 recipes with an 80/20 train/test split and regularised the model with a maximum of 2,000 iterations.
- The model chosen for its interpretability achieved an accuracy of approximately 76%.

Ingredient-Nutrition Correlation Analysis (nutritional impact of specific food items):

- Computed the Pearson correlation between each binary ingredient column and the continuous nutritional features (Calories, Fat, Sugar, Sodium, Protein, Saturated Fat).
- Filtered for significance by only including ingredients with an absolute correlation > 0.1

K-Means Clustering (To identify "Taste Clusters"):

- To group recipes based on ingredient similarity to improve recommendations
- The Silhouette score recommended k=2, but the resulting clusters lacked meaning.
- This failure revealed that the high dimensionality of ingredients creates sparse data that confuses standard distance-based clustering.

Linear Regression (User Ranking Prediction):

- For a specific user, we isolated the recipes they had rated and trained a regression model on the one-hot encoded ingredients of those specific recipes.
- Then applied this personalised model to the larger unrated dataset to predict a "User Score" for new recipes.

Random Forest & Stochastic Gradient Descent (Rating Prediction):

- To predict ratings for recipes with no user feedback, we trained Random Forest and SGD Regressor models.
- Feature Set: We expanded our features to include interaction terms like minutes_x_calories and salt_x_sugar to capture complex relationships. We also employed TF-IDF Vectorisation on the ingredient lists to create weighted features rather than simple binary counts.

Individual work (Neha Sakuja)

Dataset Research: I researched and selected the dataset. My argument was that I believe we should analyse a dataset that, along with helping us find some insights that would have real-world implications, the data shouldn't feel too alien for anyone in the team to work on. Food is a topic that everyone can relate to, irrespective of their education level.

Research Question and SH Analysis: I formulated the 1st central research question of our analysis about the Semantic Gap. I kept in mind the Business stakeholders. The retail giants and their e-retail apps have a huge influence on what consumers buy when it comes to food-related shopping.

Exploratory Data Analysis: I did a lot of EDA to understand the underlying structure of the data, to test our initial hypotheses and address some of the questions that we brainstormed before. My analysis focused on four key areas:

- Class Imbalance Analysis: I visualised the distribution of recipes across the health labels (Healthy, Neutral, Unhealthy). My analysis revealed a highly unbalanced dataset, with the vast majority of user-generated content falling into the "Unhealthy" category. This highlighted the necessity for an algorithmic intervention to bring to the surface the scarce "Healthy" options.
- Ingredient Profiling: Using frequency counts, I discovered that Salt, Butter, and Sugar were the most used ingredients. This finding showed us why users face difficulty in finding healthy options.
- I tested three common assumptions that act as barriers to healthy eating
  - Time: I compared the cooking times of healthy vs. unhealthy recipes. Contrary to popular belief, my healthy recipes do not take longer to cook.
  - Complexity: I analysed step counts and found that healthy recipes are not more complex to prepare.
  - Taste: I performed a correlation analysis between health_label and avg_rating. The results showed that healthiness has no impact on user ratings.
- Semantic Gap Analysis: I quantified the linguistic disparity between health categories. By calculating the frequency of keywords such as creamy, cheesy and rich, I discovered that unhealthy recipes are 3 to 5 times more likely to use these adjectives that people relate with comfort food. This confirmed our hypothesis that healthy food suffers from a branding disadvantage.

Key Insights & Recommendations for Business SH: In the end, I strategised the key insights for the business stakeholders and gave recommendations on actions they can take to create a large impact by changing their search recommendations, etc. and ultimately nudging the consumers to buy healthier products from their websites.

Team Collaboration: I scheduled all our team meetings, ensured that there was a constant flow of information among the team members. I structured and beautified the slides and ensured our presentation was cohesive.

Results & Conclusion

Results: We found three really important results:

1. **The Semantic Gap:** Our analysis confirmed a significant disparity in the way recipes are described. Unhealthy recipes are 3-5 times more likely to have comfort food-sounding adjectives such as rich, creamy and cheesy. This creates a psychological disadvantage for healthy options. The healthy recipes are often described using terms that may sound unappetizing to your ear. In this case, a user wouldn't even review the ingredients of a healthy recipe; maybe it does contain cheese, but you would never know.
2. **Limits of Personalisation:** We attempted to predict individual user preferences using Linear Regression and K-Means clustering, but found these methods infeasible.

- Inconsistent User Behaviour: User ratings often defy logical flavour profiles. For instance, User 145223 gave low ratings to “Mo’s Meatloaf” and “Polish Cabbage Casserole”, but high ratings to “Salisbury Steak”, dishes with very similar flavour profiles.
- Tagging Glitches: The regression model often recommended recipes based on improper tags.

3. **Predictive Modelling:** Our Random Forest and Stochastic Gradient Descent models revealed that clustering based solely on ingredients is not effective for recommendations. We concluded that recipe quality cannot be found from tags or ingredient lists alone. User ratings, therefore, remain the only reliable ground truth for quality.

Real World Implication

1. **For Retailers**: There is a business opportunity along with a chance to create a positive impact. The semantic gap shows that healthy food is suffering from a branding problem and not really a product problem. Retailers should re-tag healthy inventory with comfort keywords to boost their user engagement. This will also inadvertently give a little nudge to users to buy healthier options.
2. **For Public Health orgs**: As we found out, users respond to appetising language, the interventions of public health orgs should focus on renaming and rebranding healthy options to compete with the creamy and cheesy allure of unhealthy food.
3. **For users of Food.com:** If users rely on low-fat filters, they are likely to encounter high-sugar recipes. Users should shift their search from low-calorie to high-protein.

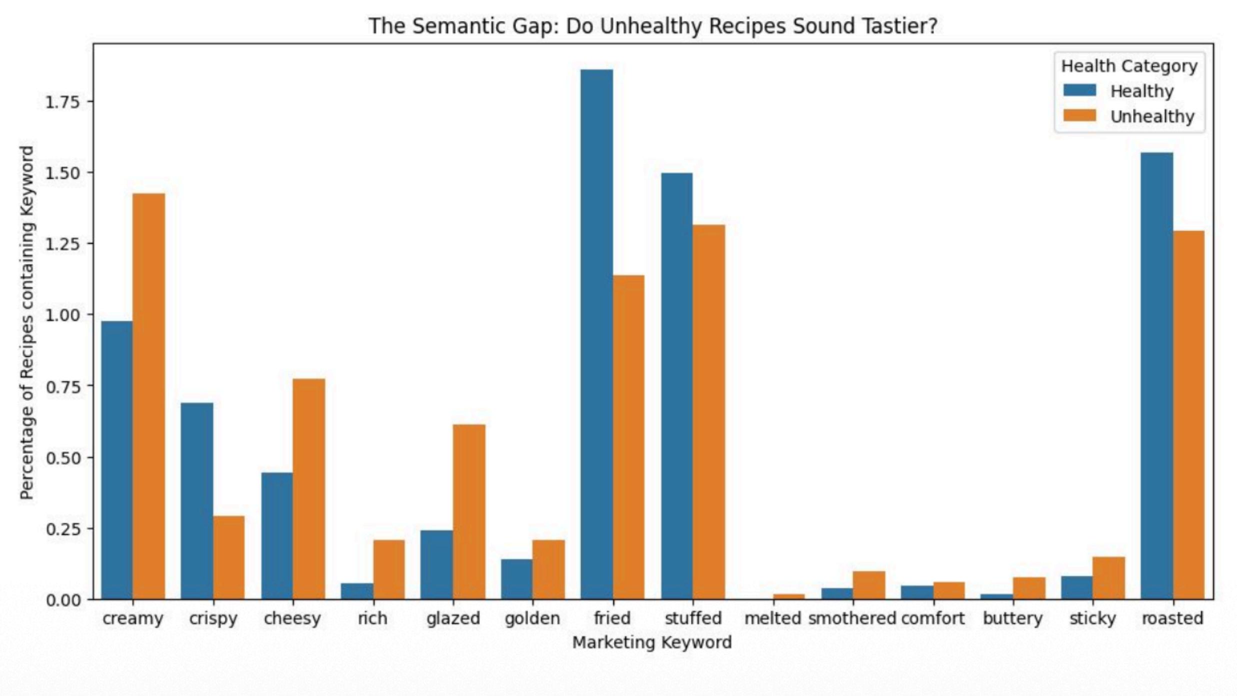
Conclusion: Our project aimed to balance health & taste through algorithmic recommendation. However, our error analysis revealed that hyper-personalisation will not work on user-generated datasets. We, therefore, conclude that simplification is superior to complexity in this domain. The most effective recommendation algorithm is not a personalised regression model, but a Bayesian Average sort injected with randomised sampling for variety. And for future work, we propose limiting recommendations based on session data, for example, if a user is on the “Quick & Easy” page, filter for low cook times. This will ensure that we provide high-quality, statistically validated healthy options rather than personalised guesses that may fail.

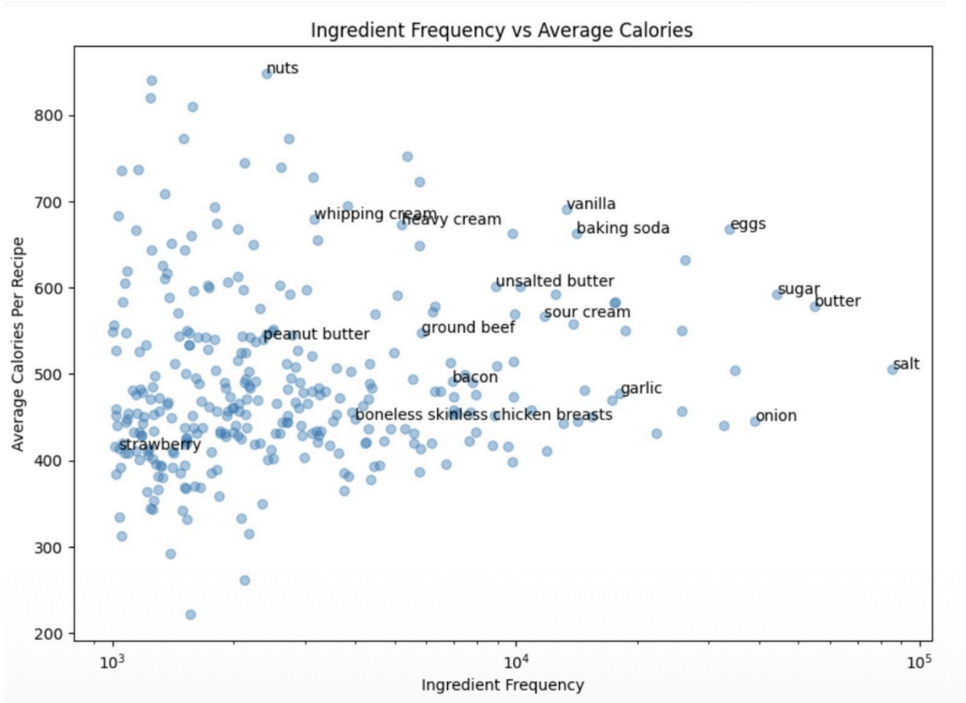
Appendix

Sources:

- Dataset: https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions/data?select=RAW_recipes.csv
- U.S. Food and Drug Administration, 03-05-2024, "What's on the Nutrition Facts Label", https://www.fda.gov/food/nutrition-facts-label/whats-nutrition-facts-label
- Office of Disease Prevention and Health Promotion, https://www.dietaryguidelines.gov/sites/default/files/2021-11/DGA_FactSheet_SaturatedFats-07-09_508c_0.pdf
- The Nutrition Source, April 2022, Harvard T.H. Chan, "Added Sugar", https://nutritionsource.hsph.harvard.edu/carbohydrates/added-sugar-in-the-diet/
- https://www.food.com/

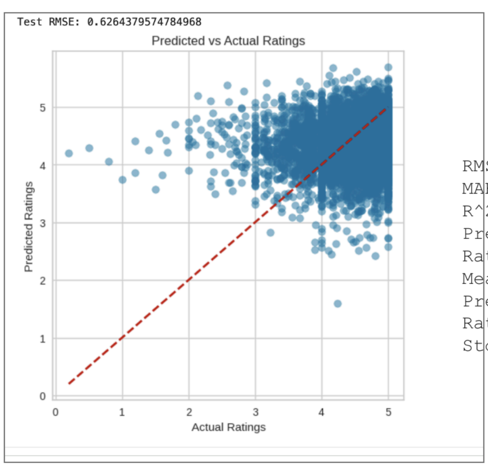
Graphs:





SGD

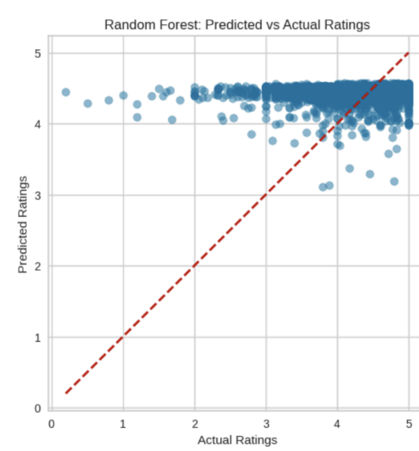
Without combined features



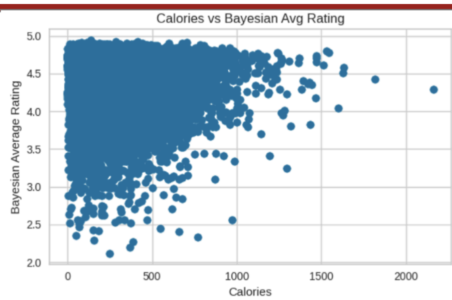
RMSE: 0.626
MAE: 0.482
R^2: -0.627
Predicted Ratings Mean: 4.359
Predicted Ratings Std: 0.405

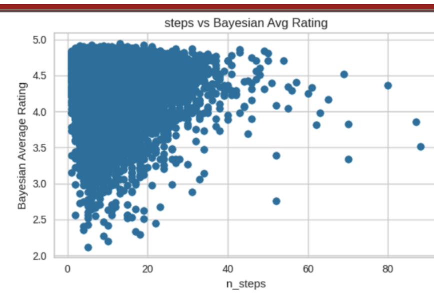
Random Forest

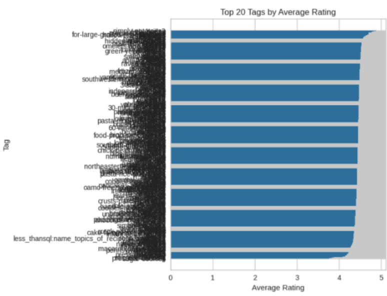
Without Combined Features

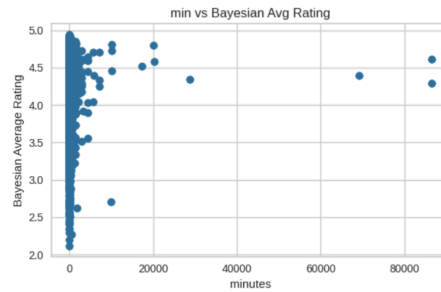


RMSE: 0.486
MAE: 0.367
R^2: 0.020
Predicted Ratings Mean: 4.454
Predicted Ratings Std: 0.088


Calories vs Bayesian Avg Rating
Bayesian Average Rating
Calories


steps vs Bayesian Avg Rating
Bayesian Average Rating
n_steps


Top 20 Tags by Average Rating
Tag
Average Rating


min vs Bayesian Avg Rating
Bayesian Average Rating
minutes